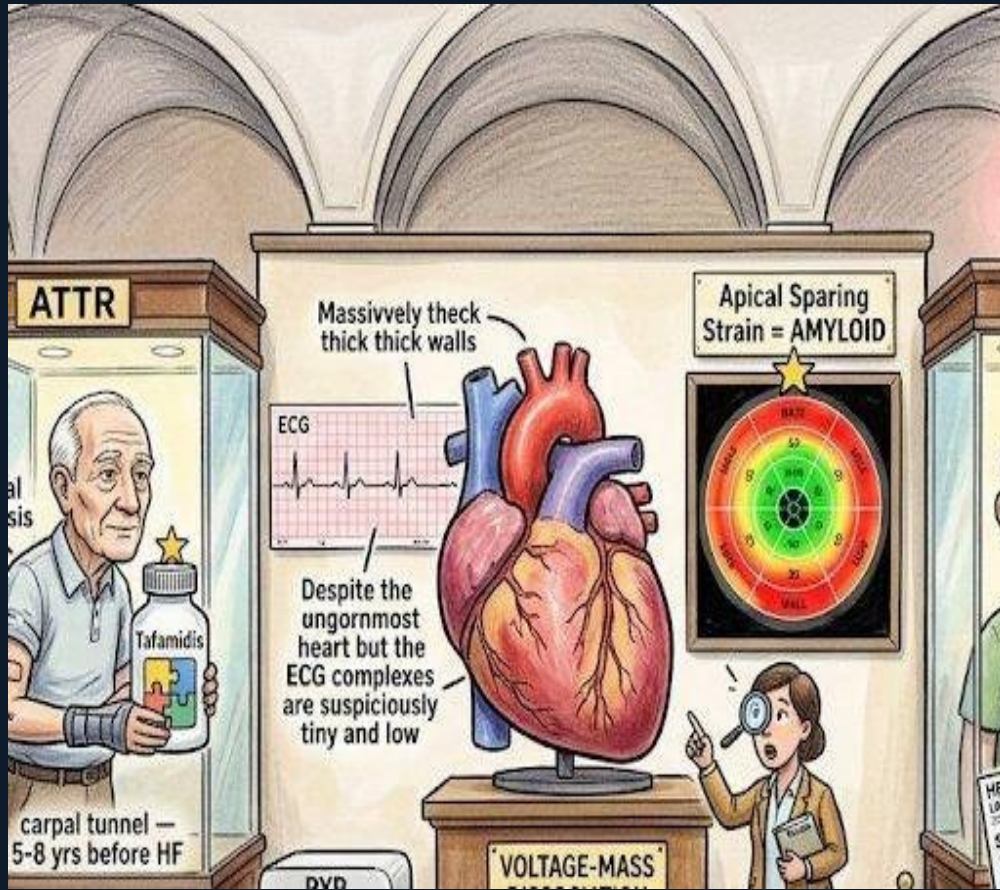
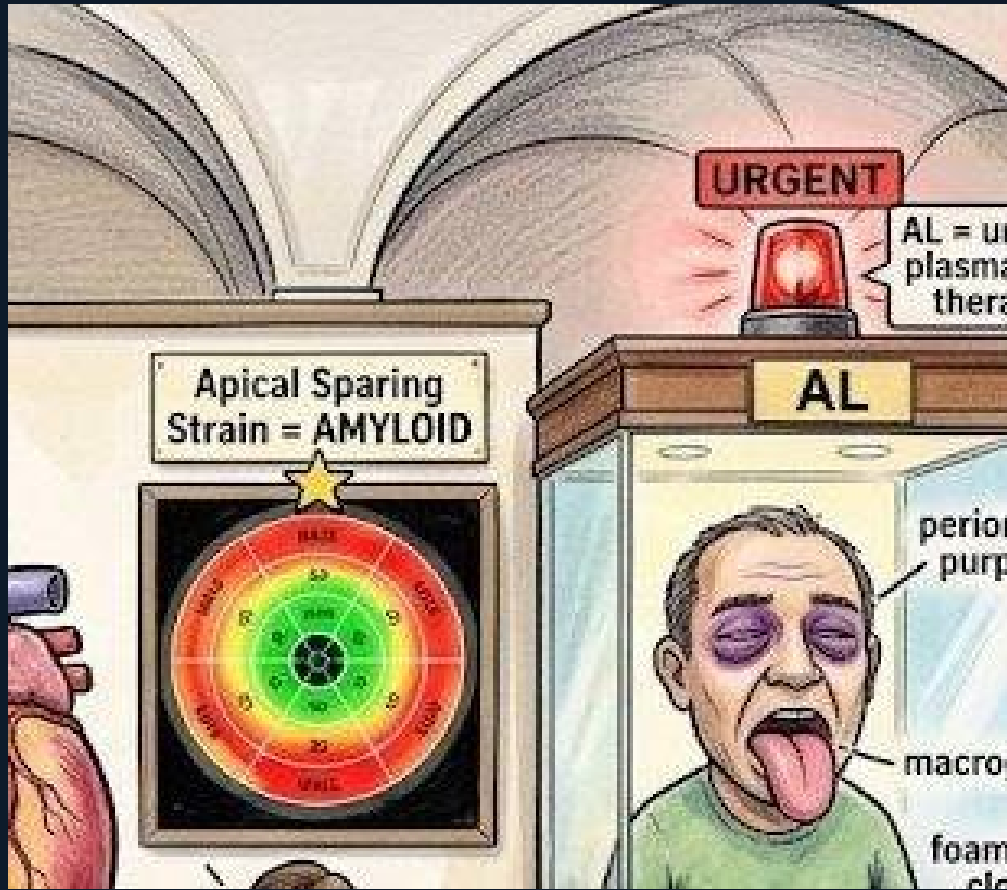




Scene D — Cardiac Amyloidosis: ATTR vs AL | Natural History Museum

Voltage-Mass Dissociation

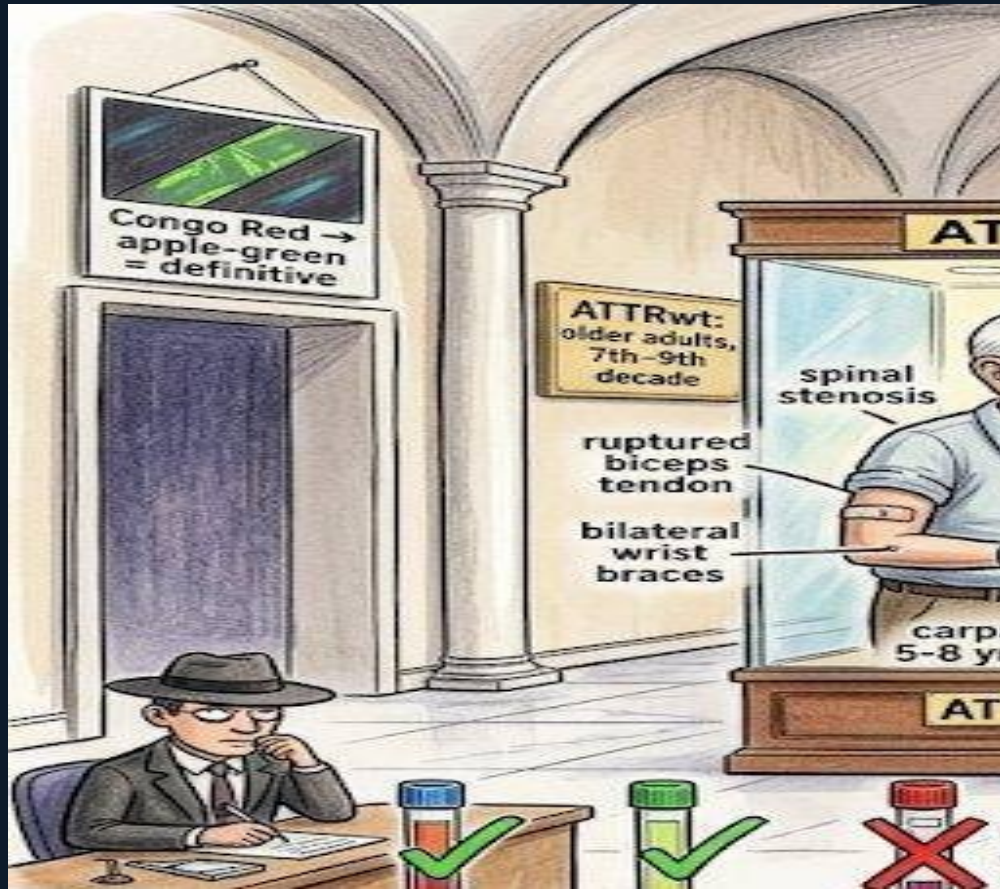
The museum guide points between two exhibits: an enormous heart with massively thick walls and an ECG strip with suspiciously tiny complexes. In HCM and hypertensive LVH, a thick wall generates high voltage — LVH criteria appear. In cardiac amyloidosis, amyloid infiltrates the extracellular space and does not conduct electricity. Result: massive wall thickness on echo but LOW or pseudo-normal voltage on ECG. This dissociation is the most specific clinical clue for cardiac amyloidosis. 'Thick LV + low voltage = amyloid' is a mandatory board rule.



Scene D — Cardiac Amyloidosis: ATTR vs AL | Natural History Museum

Apical Sparing Strain = Amyloid ★

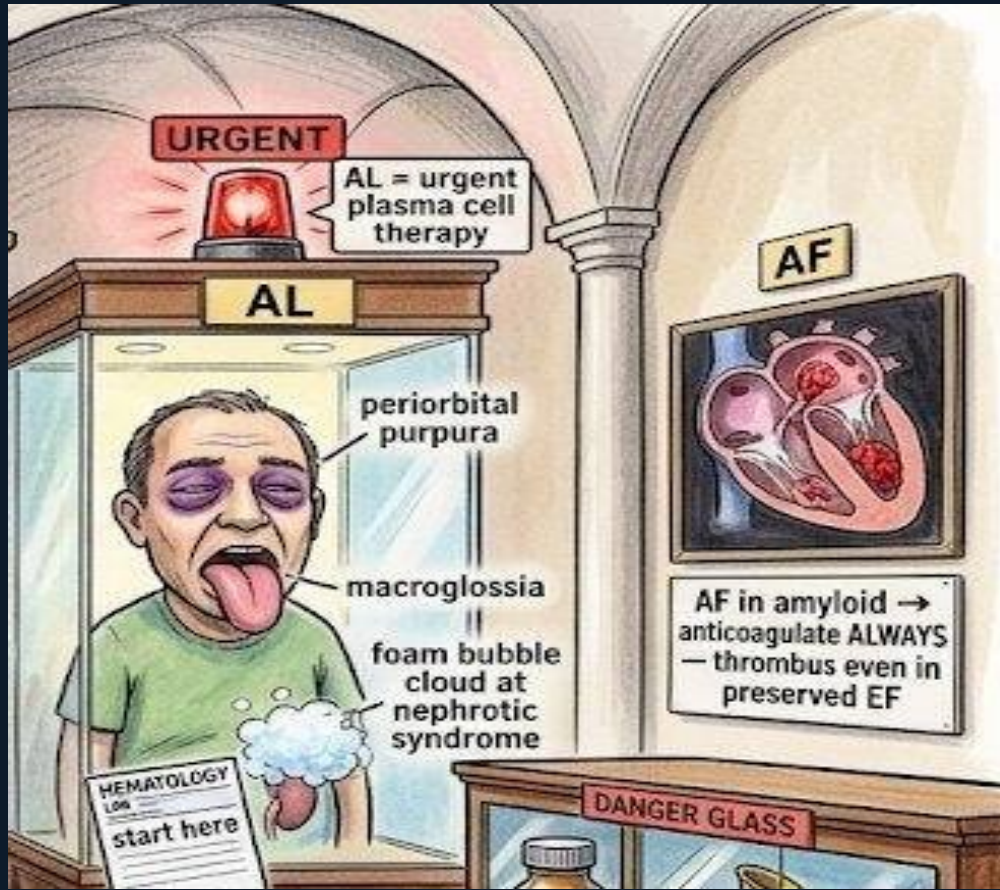
The bull's-eye pattern with normal apex (green) and abnormal base/mid-walls (red/orange) is pathognomonic for cardiac amyloidosis — gold star. Amyloid deposits are denser at the base and mid-wall; the thin-walled apex is relatively spared. This is the opposite of ischemic disease. Board rule: apical sparing on longitudinal strain = amyloid until proven otherwise. No other cardiomyopathy consistently produces this pattern. It is the single best echocardiographic discriminator for amyloid.



Scene D — Cardiac Amyloidosis: ATTR vs AL | Natural History Museum

ATTR Clues: Bilateral Carpal Tunnel

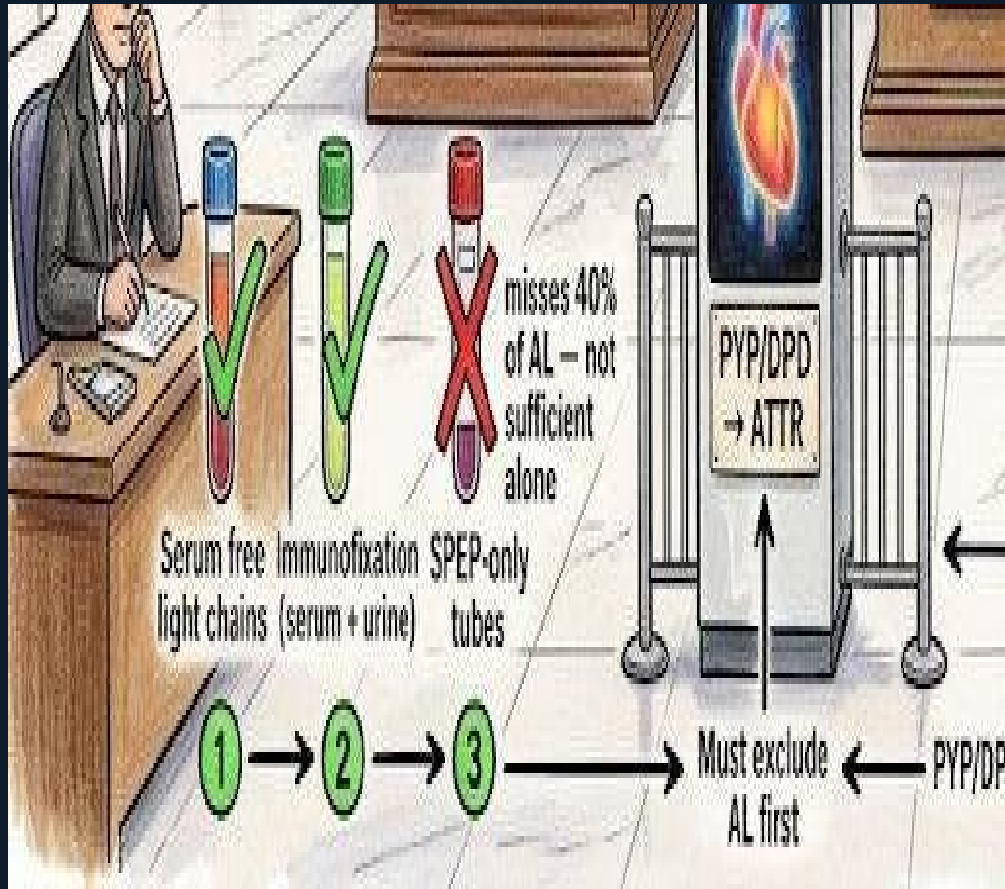
The elderly gentleman with bilateral wrist braces: bilateral carpal tunnel syndrome may precede heart failure by 5–8 years in ATTRwt. Other musculoskeletal clues: spinal stenosis, ruptured biceps tendon (Popeye deformity). Tafamidis stabilizes the TTR tetramer, preventing misfolding — disease-modifying treatment for ATTR. ATTRwt (wild-type): predominantly older men, 7th–9th decade. The board tests: 'An 80-year-old man with HFpEF, bilateral carpal tunnel, and low voltage on ECG — what is the diagnosis?' ATTRwt cardiac amyloidosis.



Scene D — Cardiac Amyloidosis: ATTR vs AL | Natural History Museum

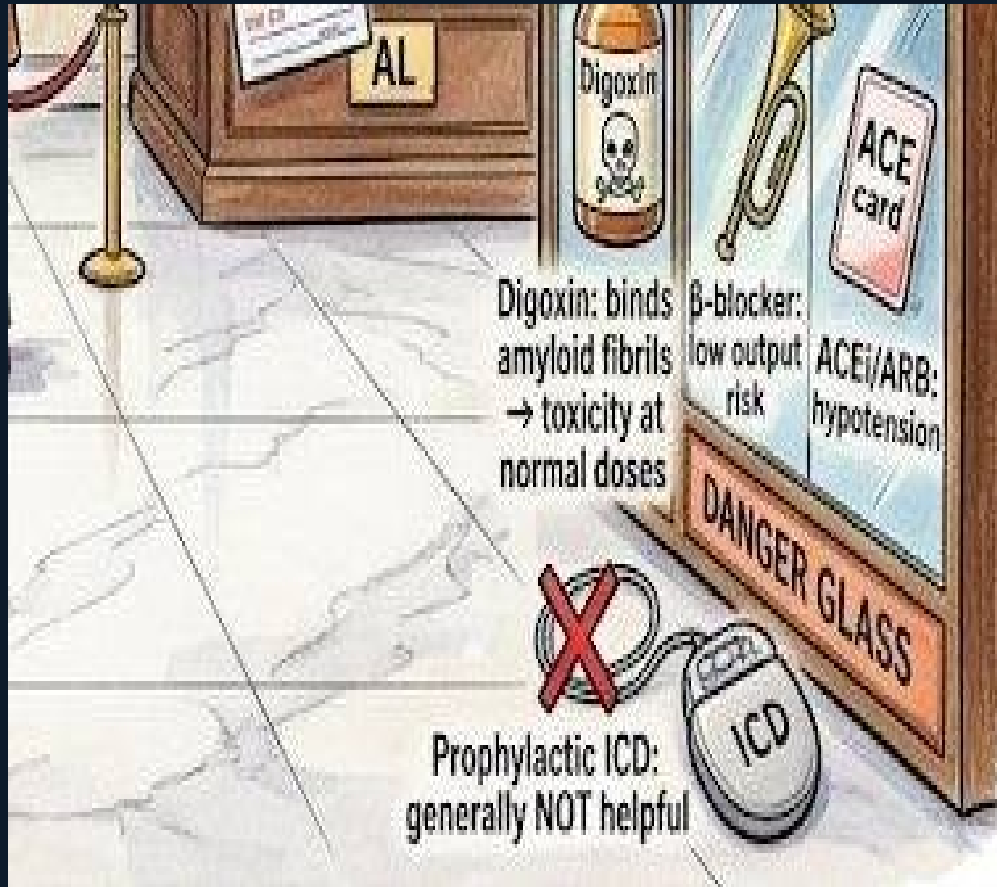
AL Amyloidosis: URGENT

Periorbital purpura (amyloid infiltration of periorbital capillaries — nearly pathognomonic for AL), macroglossia, nephrotic syndrome. AL arises from plasma cell dyscrasia. The 'URGENT' sign: AL progresses faster than ATTR — median survival without treatment is months. Urgent hematologic treatment (plasma cell-directed therapy — daratumumab-based regimens) is required. Board trap: 'Patient has thick LV, apical sparing, and periorbital purpura — what must happen next?' Urgent hematology — not tafamidis (which is for ATTR, not AL).



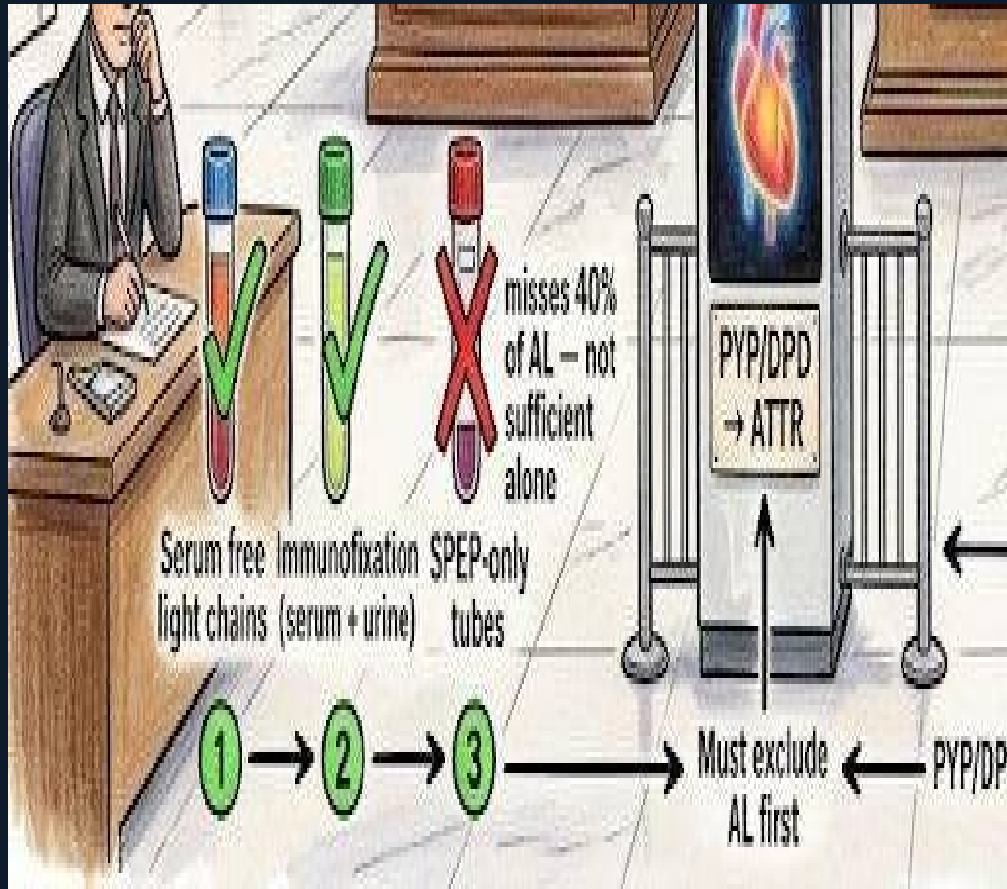
Diagnostic Sequence: Exclude AL First

Mandatory workflow: Step 1 = serum free light chains (green check). Step 2 = serum AND urine immunofixation (green check). SPEP red X: SPEP alone misses up to 40% of AL — the paraprotein may be too small to detect. After both steps negative: Step 3 = PYP/DPD bone tracer scan. Positive scan + negative plasma cell workup = ATTR — no biopsy needed. If steps 1 or 2 abnormal → hematology referral + typing. Board rule: NEVER diagnose ATTR by scan alone without first excluding AL.



Danger Cabinet + ICD Red X

Digoxin: amyloid fibrils bind digoxin → toxicity at normal therapeutic doses — avoid. Beta-blockers: amyloid hearts are stiff and rate-dependent for output; bradycardia from beta-blockade causes low output — avoid or use cautiously. ACEi/ARBs: amyloid patients have low SVR and cannot tolerate vasodilation — hypotension. Prophylactic ICD with red X: ICD is NOT routinely beneficial in amyloidosis because SCD from bradyarrhythmia or electromechanical dissociation predominates — not shockable rhythms. Consider ICD only after resuscitated arrest or sustained VT.



AF in Amyloid + Congo Red

AF in amyloid: thrombus forms even when EF is preserved because amyloid infiltrates atrial myocardium, impairing mechanical atrial contraction despite preserved LV systolic function. Anticoagulate always — standard indication regardless of $\text{CHA}_2\text{DS}_2\text{-VASc}$. Congo red slide: the definitive histologic diagnosis is Congo red staining with apple-green birefringence under polarized light. Typing by mass spectrometry distinguishes ATTR from AL when the noninvasive algorithm is inconclusive.

Scene D — Quick Reference

Thick LV + low voltage = amyloid ★

Voltage-mass dissociation — most specific clue

ATTR: bilateral CTS, spinal stenosis

May precede HF by 5-8 years

AL = URGENT — periorbital purpura

Urgent plasma cell therapy — months median survival untreated

Exclude AL first → then PYP/DPD

Positive scan + negative AL workup = ATTR

Prophylactic ICD: NOT helpful

SCD from bradyarrhythmia/EMD — not VF

Apical sparing strain = amyloid ★

Base/mid abnormal, apex spared — pathognomonic

Tafamidis = ATTR treatment

Stabilizes TTR tetramer — slows progression

SPEP misses 40% of AL

Use immunofixation + free light chains

Digoxin → toxicity in amyloid

Binds amyloid fibrils — avoid

AF in amyloid → anticoagulate always

Even preserved EF — atrial mechanical dysfunction